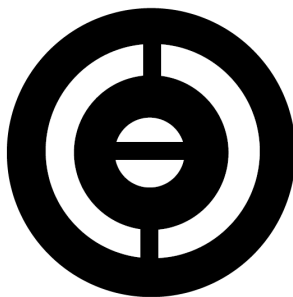




Semester II Examinations 2022-2023

Exam codes	3BA1, 4BA1, 4BME1, 1OA1, 3BMS2, 3BS91, 3FM2, 4BCT1
Exams	Third & Fourth Year & HDip
Modules	Topology
Module code	MA342 & MA535
Paper number	1
Repeat paper	No
External Examiners	Prof. Colva Roney-Dougal
Internal Examiners	Prof. Götz Pfeiffer Dr Michael Welby*
Instructions	Attempt EIGHT questions. There are ten questions, each carrying 25 marks for a total of 200 available marks.
Duration	2 hours
No. of pages	5 pages (including this one)
Departments	Mathematics
Requirements	
Release in Exam Venue	Yes ☒ No ☐
Release to Library	Yes ☒ No ☐
Statistical Tables/ Log Tables	Yes ☒ No ☐
Graph paper	Yes ☒ No ☐
Other Materials	Yes ☐ No ☒

Q1. The digital image

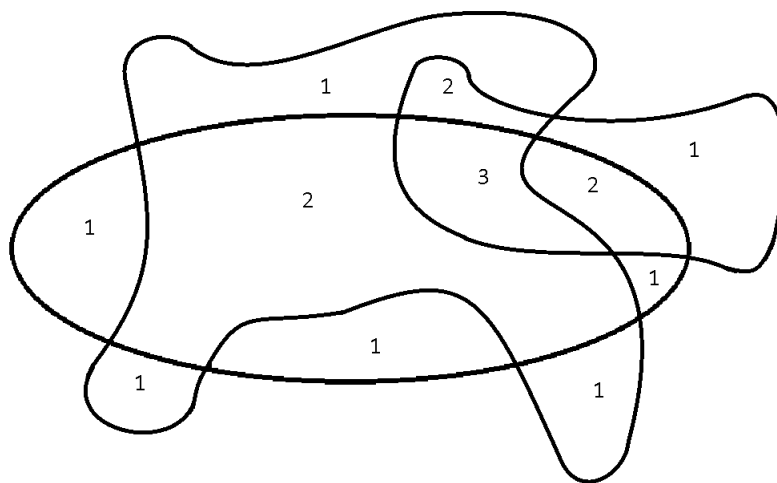


represents a closed subspace $X \subset \mathbb{E}^2$ formed as a union of many unit squares of the form $[m, m+1] \times [n, n+1]$ for integers m, n . Each such square corresponds to a tiny black pixel.

- (a) Define the Euler characteristic $\chi(X)$.
- (b) Calculate $\chi(X)$ giving details of your calculation. You may use the formula $\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B)$.

[10+15 = 25 marks]

Q2. The following picture shows the boundaries of several closed subspaces $U_1, U_2, \dots, U_t \in \mathbb{E}^2$ of common Euler characteristic $\chi(U_i) = 1$. No two boundaries are tangential.



The numbers in the interiors of the subspaces and their intersections represent a weight function $w : X \rightarrow \mathbb{Z}$ where $X = U_1 \cup U_2 \dots \cup U_t$ and $w(x) = |\{i : x \in U_i\}|$.

- (a) Define the Euler integral

$$\int_X w(x) d\chi.$$

- (b) Compute the Euler integral for this example, showing your calculations.
- (c) Derive the value of t from your calculation and state any formula you use in your derivation.

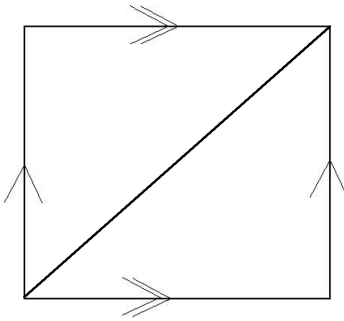
[5+10+10 = 25 marks]

Q3. For each of the following sets X and collections T of open subsets, decide if the pair (X, T) satisfies the axioms of a topological space. If the axioms are not satisfied then give one axiom that fails.

- (a) $X = \{1, 2, 3, 4, 5\}$, $T = \{\emptyset, \{1\}, \{3\}, \{1, 2\}, \{1, 2, 3\}, X\}$.
- (b) $X = \mathbb{R}^n$ and the only open subsets are X and the empty set $\emptyset$.
- (c) $X = \mathbb{Z}$ and a subset $U \subseteq \mathbb{Z}$ is open if and only if U is infinite or $\emptyset$.
- (d) $X = \mathbb{Z}$ and a subset $U \subseteq \mathbb{Z}$ is open if and only if each of its elements is even.

[6+6+8+5 = 25 marks]

- Q4.** (a) Explain what is meant by a *triangulation* of a topological space.
 (b) Explain why the following is not a valid triangulation of the torus $\mathbb{T}$.



- (c) Give a valid triangulation of $\mathbb{T}$.
- (d) Determine the number of k -simplices in your triangulation for $k = 0, 1, 2$.
- (e) Compute the Euler characteristic $\chi(\mathbb{T})$.

[5×5 = 25 marks]

- Q5.** (a) Define what it means for a topological space X to be *connected*.

For each of the topological spaces in (b)–(e), determine with justification if it is connected. If not, list the connected components.

- (b) $X = \{1, 2, 3, 4\}$ and $T = \{\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}, \{1, 2, 4\}, X\}$.
- (c) $X = \mathbb{Z}$ with the cofinite topology (a set is open if and only if it is empty or its complement is finite).
- (d) $X = \mathbb{R}^n$ with the discrete topology (all subsets of $\mathbb{R}^n$ are open).
- (e) X is the subspace of $\mathbb{E}^2$, $X = \{(x, y) \in \mathbb{E}^2 : x^2 + y^2 \neq 1\}$.

[5×5 = 25 marks]

Q6. The table

	H	M	R	C	W
H	0	5	10	8	12
M	5	0	2	11	13
R	10	2	0	12	12
C	8	11	12	0	14
W	12	13	12	14	0

gives distances between the species Human, Mouse, Rat, Cat, Whale. For $\epsilon > 0$ let G_ϵ denote the graph with vertices H, M, R, C, W and with an edge between vertices X and Y if $\text{dist}(X, Y) \leq \epsilon$.

- Sketch the graphs G_4 , G_8 , G_{10} and G_{14} .
- Draw the dendrogram that describes the inclusion relationships between the connected components of the subspaces $G_0, G_2, G_4, \dots, G_{14}$.

[4×5+5 = 25 marks]

Q7. Answer **both** (a) **and** (b).

- For topological spaces X, Y , define what it means for two continuous functions $f, g : X \rightarrow Y$ to be *homotopic*.
- Let $Y \subset \mathbb{E}^n$ be an arbitrary convex subset of Euclidean space and let X be an arbitrary topological space. Prove that any two continuous maps $f, g : X \rightarrow Y$ are homotopic.

[10+15 = 25 marks]

Q8. Answer **either** (a) **or** (b). Pay particular attention to your style of exposition.

- Define what is meant by *homotopy equivalent spaces*.
 - Prove that the circle S^1 is homotopy equivalent to the punctured complex plane $\mathbb{C} \setminus \{0\}$.
- Define what is meant by a *Hausdorff space*.
 - Prove that a compact subset A of a Hausdorff topological space X is closed. You may use the following
 - $A \subseteq V := \bigcup_{i=1}^n V_{z_k}$, where V_{z_k} is an open set containing $z_k \in A$ for $k = 1, \dots, n$.

[10+15 = 25 marks for either]

Q9. Answer **either** (a) **or** (b). Pay particular attention to your style of exposition.

- (a) State the Brouwer Fixed-Point Theorem and use it to prove the Frobenius-Perron Theorem: any real square matrix with positive entries has a positive real eigenvalue with a corresponding eigenvector having only non-negative entries.
- (b) Prove that any polynomial $a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$ with coefficients $a_i \in \mathbb{C}$ and degree $n > 0$ has at least one zero in $\mathbb{C}$. You may use the bijection $[S_1, S_1] \cong \mathbb{Z}$ that associates the homotopy class of a map with the winding number of the map.

[25 marks for either]

Q10. (a) Describe what is meant by a *mixed Nash Equilibrium*, explaining any concepts from game theory that you use.

Examine the game below. Justify your answers for (b) and (c).

- (b) Is it a pure or mixed strategy game?
- (c) Does it have a Nash equilibrium?

Two players, Player 1 and Player 2, play a game where each puts a coin on a table.

- If both coins are played with the same face up (i.e. both heads up or both tails up), then Player 1 wins €1 and Player 2 loses €1;
- If both coins are played with the different faces up (i.e. one is heads up and one is tails up), then Player 1 loses €1 and Player 2 wins €1.

The payoff table is shown below.

	H	T
H	$\nu_1(H, H) = 1$ $\nu_2(H, H) = -1$	$\nu_1(H, T) = -1$ $\nu_2(H, T) = 1$
T	$\nu_1(T, H) = -1$ $\nu_2(T, H) = 1$	$\nu_1(T, T) = 1$ $\nu_2(T, T) = -1$

[8+7+10 = 25 marks]